

# Analyzing the quality factor for Brazil

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**Abstract** Investments in Brazil are increasingly allocated to the stock market, at the expense of more conservative investments. Would finding higher-quality assets allow investors to increase their risk-return ratio? We analyze quality with several metrics, including the quality-minus-junk (QMJ) factor for Brazil. We find that quality companies are valued more by investors, with a higher price-to-book ratio. A portfolio of shares of higher quality shows a significant positive return over the period analyzed, adjusted for several risk factors. The sample members classified as quality companies remained within this classification over time.

**Keywords:** Quality minus junk; Asset pricing models; Portfolio choice; Investment decisions.

**JEL Code:** G12, G11.

## 1. Introduction

An extensive literature on risk factors explains returns on assets. One of the most famous asset pricing models, created by Fama and French (1993), has undergone several updates over the years. Asness et al. (2019) suggest a new risk factor that remunerates the investor, known as “quality minus junk” (QMJ). The intuition behind creating this factor is that higher-quality assets, all else being constant, should be more expensive than low-quality assets, eliminating higher risk-adjusted returns for a portfolio long on high-quality assets and short on assets with low-quality characteristics.

Studies show that the most profitable companies generate greater return for the investor (Novy-Marx, 2013); companies with low leverage have a higher alpha (George and Hwang, 2010; Penman et al., 2007); companies with a lower risk of credit tend to perform better; and growing companies have better returns than those that grow little (Mohanram, 2005). Asness et al. (2019) connect these disparate ideas, presenting a common characteristic among them: the quality of the company. This factor can explain a positive relationship between prices and quality. But what would a quality company

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be? For the authors, a quality company is one that grows, is highly profitable, bears low bankruptcy risks, and, hence, has a low cost of capital.

At the time of writing this paper, Brazil was going through a period with the risk-free rate at historically low levels, and negative real interest rates. Institutional investors and individuals were seeking to diversify their investments, accepting more risk. According to data from the Brazilian stock exchange (B3), the number of active accounts for individuals increased 107% in 2019, and in 2020, 92%.<sup>1</sup> We believe that studying the impact of stock picking on the return of stock portfolios becomes more relevant in such a scenario. In this study, following the suggestion of Chagas et al. (2020) to advance the relevant literature on Brazil, we apply the model proposed by Asness et al. (2019) for Brazilian stocks, to assess whether Brazilian investors are remunerated for being exposed to higher quality assets. The construction of the QMJ factor can also be used in a practical way to study the performance of funds or investors. Frazzini et al. (2018) find that the notorious investor Warren Buffet, adept at value investing, tends to buy high-quality, safe, and cheap companies. These factors may explain the significant alpha generated in his company over time.

With data collected from Brazil for the period from 2000 to 2018, our results suggest that the quality of companies is a characteristic that persists over time. That is, a company that is said to be of quality today should keep this status in later periods. In addition, we note, as do Asness et al. (2019), that the quality of the company has a positive relation to its price. Finally, we see that the QMJ factor, despite presenting high returns in the period analyzed, does not have statistical significance as a new risk factor, even when the explanation of traditional models is improved.

This paper is organized in five sections. After this introduction, we review the relevant literature, and present our methodology. We divide the results section between analysis of the persistence of quality and its impact on the price of assets and analysis of the QMJ factor. In the last section, we conclude and suggest directions for future study.

## 2. Literature review

Markowitz (1959) was one of the first scholars to consider not only returns, but also the risk involved for an investor. He divides the risk to which

<sup>1</sup> Year-to-year percentage change of personal accounts at the end of each year are available at [http://www.b3.com.br/pt\\_br/market-data-e-indices/servicos-de-dados/market-data/consultas/mercado-a-vista/historico-pessoas-fisicas/](http://www.b3.com.br/pt_br/market-data-e-indices/servicos-de-dados/market-data/consultas/mercado-a-vista/historico-pessoas-fisicas/).

the investor is exposed into two components: systematic risk, which is not diversifiable; and specific risk, which is. From this theory, Sharpe (1964) develop a model, based on the study by Tobin (1958), which is known as the capital asset pricing model (CAPM). One of the implications of the model is that a diversified investor can only achieve greater returns when exposed to a greater market risk, the sensitivity known as beta.

CAPM is still widely used in practice and in academic studies, despite evidence that market risk is not sufficient to fully explain returns on assets and portfolios. Based on the CAPM, Jensen (1968) develops a measure of the skill of a portfolio manager, which consists of the constant (alpha) of the regression equation commonly used for empirical tests of the CAPM. Several researchers find evidence of returns consistently above those postulated by the CAPM (statistically positive alpha). These pieces of evidence are commonly called anomalies. One of the oldest anomalies studied is the value factor. Graham and Dodd (1934) document that companies with value perform better than companies with lower value, known as growth companies (growth). This anomaly has been explored in depth by other authors such as Basu (1975, 1983) and Stattman (1980), who show, by different metrics, that the value factor can explain part of the returns on assets.

Banz (1981) find evidence of a negative relationship between the size of the company and its return, so that small companies have higher returns than large companies. This anomaly, called the size factor, has persisted for at least forty years. It is incorporated by Fama and French (1993), together with the value factor, and the market risk premium, into the model today known as the Fama and French three-factor model.

Jegadeesh and Titman (1993) find another anomaly: the momentum factor. According to them, it is possibly linked to the behavior of investors. The authors realize that buying companies that performed well and selling those that did not perform so well in the past generates abnormal returns for investors. One of the authors' main arguments is that the market reacts slowly to new information specific to firms, a phenomenon known as *underreaction*. The strategy of maintaining a portfolio long on winning shares (with good past performance) and short on losing shares (with low past performance) is the momentum factor. Carhart (1997) also finds evidence of the influence of the momentum factor on asset returns. He suggests a four-factor model, adding this factor to the earlier three factors of Fama and French.

Through a financing constraint model, Frazzini and Pedersen (2014) prove that a strategy of maintaining a portfolio long on low-beta assets and short on high-beta assets, known as the "betting against beta" (BAB) factor, pro-

duces a positive excess return, even with beta zero. According to the model, it presents positive returns when there is liquidity in financing, and negative returns in liquidity crises. The authors obtain positive alphas, not only for CAPM, but also for several other pricing models with more factors, in the U.S. and in nineteen other developed markets.

Some investigators discuss the idea that investors obtain abnormal returns if they maintain a strategy of being long on quality companies and short on companies considered to have low quality. But how to classify and create a quality measure? [Graham \(1973\)](#) uses metrics of indebtedness, stability of profits, and growth of dividends, together with metrics of value (price-to-book and price-earnings) to show that choosing cheap and quality companies compensates investors. [Greenblatt \(2010\)](#) defines quality as a high return on invested capital, combined with a high net profit margin. [Novy-Marx \(2013\)](#) finds that companies with high gross profitability generate excess return not explained by Carhart's four-factor model ([Carhart, 1997](#)). Based on the discounted dividend model, [Fama and French \(2015\)](#) add two new factors: profitability and investment, to the Fama and French three-factor model. They note that including these factors reduces the relevance of the value factor to explain excess returns. However, [Martins and Eid Jr \(2015\)](#) show that, in Brazil, these two new factors have low explanatory power.

[Asness et al. \(2019\)](#) combine several variables in a z-score methodology based on [Gordon's \(1962\)](#) growth model to create the factor known as "quality minus junk" (QMJ). The authors realize that the quality characteristic is not fully priced in the actions. They discuss whether this positive adjusted return can come from human behavior, regulatory constraints, or the wrong specification of risk models. The excess return persists in the North American data, and is also observed in twenty-three more countries. In Brazil, as far as we are aware, this factor has not been tested. It was suggested as a topic for future research by [Chagas et al. \(2020\)](#), on the possibility of gains from a well-founded stock portfolio.

### 3. Methodology and Database

We initially defined the universe of all companies listed on the São Paulo stock exchange. For each company, we chose the main stock, based on Bloomberg's classification of primary stocks. In all, we collected a sample of 355 shares listed between 2000 and 2018. Also, in each month, for a share to be eligible for our analysis, it must have had an average volume greater than 500 thousand Brazilian Reais (BRL) per day, on more than 80% of trading days during the previous year. These filters follow those used by the Center

for Research in Financial Economics (NEFIN) at the University of São Paulo (USP) in its studies related to risk factors. Thus, we guarantee that we are collecting a universe of shares available over the period analyzed, including shares that are no longer listed, and a universe that can be traded without major problems.

We collected balance sheet data, as well as price and market value data, from Bloomberg. We calculated returns in BRL, and adjusted them for dividends and other corporate events. We calculated excess returns on the 30-day *depósito interbancário* (DI) swap, which is a future-interest-rate contract traded on B3. It is more liquid than the government bonds secondary market, and can be considered to be risk-free. Minardi and Sanvicente (2006) discuss alternative representations of the risk-free interest rate to estimate the cost of capital in Brazil. To this end, they conduct their analysis with savings accounts and DI. They note that the correlation between them is 0.9. Therefore, we see that our analysis would not change if we used the DI as a risk-free asset, instead of savings accounts or government bonds, for instance.

Similar to Asness et al. (2019), our main object of study, the quality factor, is based on Gordon's growth model. In it, price is defined as the dividend flow divided by a rate given as the difference between the rate of return required by the company and its growth. Since price is not a stationary variable over time, we use a measure of relative value, the price-to-book value (PB). Thus, we obtain the following definition of price behavior:<sup>2</sup>

$$PB = \frac{\text{Profitability}}{\text{Safety} - \text{Growth}}. \quad (1)$$

Based on this model, it is possible to define what makes a company trade at a higher price. We consider a company to have quality if it is profitable, has growth, and is safe. Creating these variables followed the technique used by Asness et al. (2019). A standardization methodology (z-score) combined each indicator, transforming the measures into deviations from the mean. Essentially, we ranked the indicators and then subtracted each value from the cross-sectional average. Finally, we divided them by the standard deviation, also cross sectionally:

$$z_x = \frac{r - \mu_r}{\sigma_r},$$

<sup>2</sup>By rewriting the Gordon model we obtain

$$PB = \frac{1}{B} \frac{\text{Dividend}}{\text{required return} - \text{growth}} = \frac{(\text{Profit}/B) \times (\text{Dividend}/\text{Profit})}{\text{required return} - \text{growth}} = \frac{\text{Profitability} \times \text{Payout ratio}}{\text{required return} - \text{growth}}.$$

The *payout ratio* measure, despite having been studied by Asness et al. (2018) in their 2014 article, was removed in the 2018 article. The authors changed the name of *Required Return* to *Safety* because the safer a company is, the lower the required return.

where  $r$  is the vector of the  $x$  variable ranked;  $\mu_r$  is the cross-sectional mean of  $r$ ; and  $\sigma_r$  is the cross-sectional standard deviation of  $r$ .

### 3.1 Components

We describe each variable presented in Equation (1) below:

1. *Profitability*: It combines gross profit on total assets (GPOA), return on equity (ROE), return on assets (ROA), cash flow on total assets (CFOA), gross margin (GMAR), and accruals (*accruals*, ACC):

$$\text{Profitability} = z(z_{\text{GPOA}} + z_{\text{ROE}} + z_{\text{ROA}} + z_{\text{CFOA}} + z_{\text{GMAR}} + z_{\text{ACC}})$$

2. *Growth*: It combines variations of all *Profitability* measures, with the measures transformed into units per share. It omits the variation in provisions derived from *accruals*. We look at the growth of variables over the past 5 years:

$$\text{Growth} = z(z_{\Delta\text{GPOA}} + z_{\Delta\text{ROE}} + z_{\Delta\text{ROA}} + z_{\Delta\text{CFOA}} + z_{\Delta\text{GMAR}})$$

3. *Safety*: It consists of the opposite of the betas, estimated against the Ibovespa index (BETA). The lower the beta, the greater the contribution to *safety*. Additionally, this measure has a leverage indicator measured as the company's total debt (LEVERAGE), two measures of bankruptcy risk (Ohson's O and Altman's Z), and the volatility of return on equity (ROE):

$$\text{Safety} = z(z_{\text{Beta}} + z_{\text{LEVERAGE}} + z_{\text{O-SCORE}} + z_{\text{Z-ALTMAN}} + z_{\text{ROE VOL}})$$

Finally, we add all these variables together and re-apply the z-score to obtain the final quality indicator:

$$\text{Quality} = z(\text{Profitability} + \text{Growth} + \text{Safety}).$$

After applying the filters described above, we have an average of 239 listed companies per year, with an average size of almost 4 billion BRL, a relatively low value. Just as an example, currently, a company like Ambev, considered one of the largest Brazilian companies, has a market value of approximately 270 billion BRL. A small company, Bombril, popularly known for its steel wool, has only 450 million BRL of market value.

The construction of the *quality minus junk* factor—hereinafter expressed as QMJ—follows the methodology of [Asness et al. \(2019\)](#). We first divide

companies by size, defined as market capitalization. For each month, we separate the 20% largest companies (*Big*) and the 20% smallest companies (*Small*). Within each subgroup, we rank companies according to their quality score. Finally, we divide them into two groups, classified as *Quality* and *Junk*. Therefore, the *Small Quality* portfolio, for instance, comprise the 20% smaller companies, among which 50% have higher quality.

The QMJ risk factor is, therefore, the average returns of the two portfolios with high-quality companies, less the average returns of the two portfolios with lower quality:

$$\begin{aligned} \text{QMJ} &= \frac{1}{2} (\text{Small Quality} + \text{Big Quality}) - \frac{1}{2} (\text{Small Junk} + \text{Big Junk}) \\ &= \frac{1}{2} (\text{Small Quality} - \text{Small Junk}) + \frac{1}{2} (\text{Big Quality} - \text{Big Junk}). \end{aligned}$$

The weights given to the portfolio assets are relative to their market capitalizations. Therefore, given a period  $t$ , expressed in months, asset  $i$  has the following weight:

$$w_{i,t} = \frac{\text{Market Cap}_{i,t}}{\sum \text{Market Cap}_{i,t}}. \quad (2)$$

Following this same methodology, we build *long / short* portfolios for individual quality components (*profitability*, *growth*, and *safety*). We use NEFIN data for the other risk factors, namely market, momentum, size, and value. The factors created by NEFIN follow eligibility criteria similar to those created in this study, where a stock traded on B3 is considered eligible if it meets minimum trading and listing criteria over time.

## 4. Results

Table A1 shows a statistical summary of the data used to calculate the quality metric. In this table we can see some outliers. Most of these companies suffered corporate events that impacted their published data. Since the construction of the indicators uses z-score methodology of the ranking, rather than absolute value, these outliers are not a concern.

### 4.1 Persistence of quality

In this section, we examine whether the attribute “quality” is persistent over time. That is, if a company is classified as part of the group of quality companies, will it still be classified in this same group in the future? To do this, we analyze the difference in the z-score of the highest-quality companies

(1st decile) with that of the lowest-quality firms (10th decile). If the difference is statistically positive, even after 24 months of classification, we consider that the attribute is persistent. Table A2 shows the result of the persistence analysis of the quality score indicator. For this test, we rank companies on a monthly basis by their quality score. We allocate them to one of the ten portfolios, with the first portfolio containing the highest-quality companies and the tenth portfolio containing the lowest-quality companies. Having done that, we analyze these portfolios, weighted by the size of the companies, at time  $t$  (portfolio formation) and in the next 3, 6, 12 and 24 months ( $t + 3, 6, 12,$  and 24 months). It is worth noting that, unlike Asness et al. (2019), we do not perform this analysis for a window of 120 months, due to insufficient data. The table therefore shows the average of the time series for each portfolio, with standard errors adjusted for heteroskedasticity and autocorrelation with a 6-month lag (Newey and West, 1987).

Based on the results in Table A2, and verified by the p-values of the difference between z-scores of the high- and low-quality companies, after 24 months we still manage to reject the null hypothesis of equality between them. On average, quality companies retain this characteristic in all windows. Despite the score falling over time, the order of the portfolios is maintained. Thus, we find persistence in the quality attribute.

Looking at the quality measures individually (*Profitability*, *Growth*, and *Safety*), we continue to reject the null hypothesis of an equal score between the portfolios, despite observing a small change in order in the Growth measure after 24 months.

Therefore, we can conclude that quality is a persistent characteristic. This suggests that higher-quality companies today are more likely to continue to have this characteristic in the future. The individual measures of quality (*Profitability*, *Growth*, and *Safety*) behave the same as the aggregate component, unlike what Asness et al. (2019) finds, which is less persistence of individual measures, compared to the aggregate component.

## 4.2 The price of quality

To analyze the influence of quality on the price of shares of companies, we perform a *cross-sectional* regression of the logarithm of the *price-to-book value* (PB). The regression can be expressed as follows:

$$P_t^i = a^i + b^i \text{Quality}_t^i + \varepsilon_t^i, \quad (3)$$

where  $P_t^i = \log(\text{PB})_t^i$  is the logarithm of the *price-to-book* of company  $i$  at time  $t$ ;  $\text{PB} = \text{Profitability}/(\text{Safety} - \text{Growth})$ ; and  $\text{Quality}_t^i$  is the z-score of

company  $i$  at time  $t$ .

Interpreting parameter  $b^i$  is simple: it measures the percentage increase in the *price-to-book* (PB), given an increase of a standard deviation in the quality score. Asness et al. (2019) show a positive relationship between price and quality. The coefficients are positive and significant in the United States and in the global sample of countries. For the Brazilian market, the data show very similar results. Table A3 shows the results from a sample that runs from 2000 to 2018 for the cross-section regression of Fama and MacBeth (1973). That is, the dependent variable is the log of the equity value price of shares, and the independent variables are the quality z-score and its individual components. The beta is 0.1741, with a t-statistic of 18.55, and an adjusted average  $R^2$  of 2.7%. As highlighted, the result is quite similar to that found by Asness et al. (2019). In the Brazilian market, the price of shares correlates positively with their quality.

We also analyze the individual components of the z-score using the following equations:

$$P_t^i = a^i + b^{i,1} \text{Profitability}_t^i + \varepsilon_t^i \quad (4)$$

$$P_t^i = a^i + b^{i,2} \text{Growth}_t^i + \varepsilon_t^i \quad (5)$$

$$P_t^i = a^i + b^{i,3} \text{Safety}_t^i + \varepsilon_t^i \quad (6)$$

$$P_t^i = a^i + b^{i,1} \text{Profitability}_t^i + b^{i,2} \text{Growth}_t^i + b^{i,3} \text{Safety}_t^i + \varepsilon_t^i \quad (7)$$

Table A3 shows that all measures, seen individually, have statistically-significant coefficients. *Profitability* and *Growth* show positive coefficients, as expected, while *Safety* has a negative coefficient. Equation 7 shows an improvement in explanatory power from 2.7% to almost 9%, also observed by Asness et al. (2019). Our results are partially consistent with others in the literature, and with the hypothesis that higher-quality companies demand higher prices from investors, observed by the aggregate measure. Among the individual measures, *Profitability* stands out, consistent with the literature in both simple and multiple regressions. Despite the statistically significant coefficients, the effect on price of both *Growth* and *Safety*, individually, is very low. The explanatory power of regressions using only these measures is also very low. The divergent result of the safety measure indicates a small and negative influence on the stock price. That is, safe companies (with low systemic risk, low risk of bankruptcy, and low volatility in profitability) have a relatively lower price in the sample. The growth variable results are inconclusive.

### 4.3 Quality as a risk factor

Table A4 shows correlations between risk factors. We see that QMJ has a positive relationship with its individual components. *Growth* is the variable with the weakest relationship with QMJ (below 0.2). *Growth* and *Safety* have a negative, although low, correlation with each other. Table A4 also shows the correlation of these factors with more traditional factors, such as the market factor, represented as MKT; the *Value* and *Size* factors of Fama and French (1993), represented as HML and SMB; and the *Momentum* factor, represented as UMD. QMJ has a negative relationship with these; only *Momentum* has a positive correlation. Considering the individual components of the quality score, there is a very similar relationship. The exception is *Growth*, which has almost no correlation with the main factors.

Figure A1 reports the performance of risk factors over the period analyzed. We note that, overall, the factor had an excess of positive return, but it presents some moments of very negative performance.

The figure shows the accumulated excess return of quality minus junk (QMJ), high minus low (HML), up minus down (UMD), small minus big (SMB), and market (MKT) factors. The period analyzed runs from 2001 to 2018.

### 4.4 Excess returns from portfolios with quality assets

Table A5 shows the return of the shares, ordered in 10 deciles by the quality score. The first decile has the return of portfolios with lower-quality assets, and the tenth decile the return of portfolios with higher-quality assets. The first line of the table shows the excess return (on the risk-free asset). In the other lines the alphas relate to the estimates in Equations (8) to (11):

$$R_t - R_{f_t} = \alpha^{\text{CAPM}} + \beta^{\text{MKT}} \text{MKT}_t + \varepsilon_t \quad (8)$$

$$R_t - R_{f_t} = \alpha^{\text{FF3}} + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{SMB}} \text{SMB}_t + \beta^{\text{HML}} \text{HML}_t + \varepsilon_t \quad (9)$$

$$R_t - R_{f_t} = \alpha^{\text{CARHART}} + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{SMB}} \text{SMB}_t + \beta^{\text{HML}} \text{HML}_t + \beta^{\text{UMD}} \text{UMD}_t + \varepsilon_t \quad (10)$$

$$R_t - R_{f_t} = \alpha^{\text{5 Factor}} + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{SMB}} \text{SMB}_t + \beta^{\text{HML}} \text{HML}_t + \beta^{\text{UMD}} \text{UMD}_t + \beta^{\text{QMJ}} \text{QMJ}_t + \varepsilon_t. \quad (11)$$

Results in Table A5 show that the excess return gradually increases with

the quality level of the portfolios, with positive excess returns and less systematic risk. This shows that Brazilian investors benefit from maintaining quality companies in their portfolio, a result that aligns with [Asness et al. \(2019\)](#). It appears that the intercepts of all tested functional forms are positive and statistically significant for the portfolios of higher quality (last decile). For instance, a market factor alpha of 0.15 suggests that the high-quality portfolio earned 15% more than the return required due to the market risk taken over the year. This suggests that by adding a new unexplored risk factor to the Brazilian market, we can explain part of the cross-sectional variability.

#### 4.4.1 Behavior of other factors with the addition of the *Quality Minus Junk* factor

Table A6 is used to verify the effect of adding the QMJ factor on the alphas and betas of the regressions of commonly known factors. [Varga and Brito \(2016\)](#), with Brazilian data from 1999 to 2015, show that HML and UMD account for part of the cross-sectional returns, while the market factor and SMB do not play an important role. With the data used in this article, we find something very similar for the equations without QMJ. Table A6 reports the results of the following regressions, including and excluding the QMJ factor for each regression:

$$\text{SMB}_t = \alpha + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{HML}} \text{HML}_t + \beta^{\text{UMD}} \text{UMD}_t + \beta^{\text{QMJ}} \text{QMJ}_t + \varepsilon_t \quad (12)$$

$$\text{HML}_t = \alpha + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{SMB}} \text{SMB}_t + \beta^{\text{UMD}} \text{UMD}_t + \beta^{\text{QMJ}} \text{QMJ}_t + \varepsilon_t \quad (13)$$

$$\text{UMD}_t = \alpha + \beta^{\text{MKT}} \text{MKT}_t + \beta^{\text{HML}} \text{HML}_t + \beta^{\text{SMB}} \text{SMB}_t + \beta^{\text{QMJ}} \text{QMJ}_t + \varepsilon_t. \quad (14)$$

There is no statistical evidence of an alpha other than 0 for the SMB factor when controlled by the MKT, UMD, and HML factors, even after adding the QMJ factor, unlike the results of [Asness et al. \(2019\)](#). However, we find evidence similar to theirs of a negative beta between QMJ and SMB. This is intuitive, since larger companies are generally considered to be safer, with more constant growth, that is, higher quality. The HML factor shows a positive annualized return, but only has statistical significance after adding the QMJ factor as an explanatory variable. In this case, the beta of the factor is

negative, as seen by [Asness et al. \(2019\)](#) in the United States and global samples. This relationship is also intuitive, and speaks to our previous analyses of price and quality.

Finally, the UMD factor is the only one that presents positive and significant performance and alpha for the Brazilian data, regardless of the presence of the QMJ factor in the regression. This relationship indicates that companies in *Momentum* are generally quality companies. We conclude that including the QMJ factor improves the explanation of the traditional factors, and hence, it could be used for studies related to this theme.

#### 4.5 Risk premium

We use the methodology of [Fama and MacBeth \(1973\)](#) to find the prize for exposure in each factor. This methodology consists of two stages. In the first stage, the portfolios / assets are regressed against the time series of the factors, to determine how much they are exposed to the factors. In the second stage, *cross-sectional* returns are regressed against the betas estimated in the first stage, thus generating a time series of the coefficients of these risk premiums.

Translating this into equations, for  $n$  assets and for the factors *Market* (MKT), *Value* (HML), *Size* (SMB), *Momentum* (UMD), and *Quality* (QMJ), in the first step, we obtain the betas through the following equations:

$$R_{1,t} - R_{f_t} = \alpha_1 + \beta_1^{\text{MKT}} \text{MKT}_t + \beta_1^{\text{SMB}} \text{SMB}_t + \beta_1^{\text{HML}} \text{HML}_t + \beta_1^{\text{UMD}} \text{UMD}_t + \beta_1^{\text{QMJ}} \text{QMJ}_t + \varepsilon_{1,t}$$

$$R_{2,t} - R_{f_t} = \alpha_2 + \beta_2^{\text{MKT}} \text{MKT}_t + \beta_2^{\text{SMB}} \text{SMB}_t + \beta_2^{\text{HML}} \text{HML}_t + \beta_2^{\text{UMD}} \text{UMD}_t + \beta_2^{\text{QMJ}} \text{QMJ}_t + \varepsilon_{2,t}$$

⋮

$$R_{n,t} - R_{f_t} = \alpha_n + \beta_n^{\text{MKT}} \text{MKT}_t + \beta_n^{\text{SMB}} \text{SMB}_t + \beta_n^{\text{HML}} \text{HML}_t + \beta_n^{\text{UMD}} \text{UMD}_t + \beta_n^{\text{QMJ}} \text{QMJ}_t + \varepsilon_{n,t}$$

$R_{i,t}$  represents the return of each asset  $i$  out of a total of  $n$ , where  $n$  is the total number of individual shares in the sample. The  $\beta$ s are what we know as the *loading factor*. They represent how much the returns on assets are exposed to each of these factors.

In the second stage of this procedure, we perform  $T$  *cross-sectional* re-

gressions using the  $\beta$ s estimated in the first stage for each factor. The objective now is to find the exposure of the returns of each company to risk factors over time. That is, the hypothesis that we test is: does greater exposure to the risk factor demand greater returns?

$$R_{i,1} - R_{f_1} = \gamma_1^0 + \gamma_1^{\text{MKT}} \hat{\beta}_i^{\text{MKT}} + \gamma_1^{\text{SMB}} \hat{\beta}_i^{\text{SMB}} + \gamma_1^{\text{HML}} \hat{\beta}_i^{\text{HML}} + \gamma_1^{\text{UMD}} \hat{\beta}_i^{\text{UMD}} + \gamma_1^{\text{QMJ}} \hat{\beta}_i^{\text{QMJ}} + \varepsilon_{i,1}$$

$$R_{i,2} - R_{f_2} = \gamma_2^0 + \gamma_2^{\text{MKT}} \hat{\beta}_i^{\text{MKT}} + \gamma_2^{\text{SMB}} \hat{\beta}_i^{\text{SMB}} + \gamma_2^{\text{HML}} \hat{\beta}_i^{\text{HML}} + \gamma_2^{\text{UMD}} \hat{\beta}_i^{\text{UMD}} + \gamma_2^{\text{QMJ}} \hat{\beta}_i^{\text{QMJ}} + \varepsilon_{i,2}$$

⋮

$$R_{i,T} - R_{f_T} = \gamma_T^0 + \gamma_T^{\text{MKT}} \hat{\beta}_i^{\text{MKT}} + \gamma_T^{\text{SMB}} \hat{\beta}_i^{\text{SMB}} + \gamma_T^{\text{HML}} \hat{\beta}_i^{\text{HML}} + \gamma_T^{\text{UMD}} \hat{\beta}_i^{\text{UMD}} + \gamma_T^{\text{QMJ}} \hat{\beta}_i^{\text{QMJ}} + \varepsilon_{i,T}.$$

In this second regression, the  $\gamma$ s are used to calculate the risk premium for each factor. Altogether, we have six time series of  $\gamma$ s, one for each factor. To find the risk premium for each one, we take the average of the time series (assuming that the errors are i.i.d.).

Table A7 shows the results of this methodology. We see that adding the risk premium for the quality factor to the Carhart model (Carhart, 1997) produces a statistically irrelevant and negative premium. We also notice that only in the Fama and French model (Fama and French, 1993), does a factor other than the market factor manage to have a statistically relevant coefficient, with a 90% confidence interval. Including QMJ in the Carhart model improves the adjusted determination coefficient, but the difference between the Carhart model and the model with QMJ is very low.

Thus, we conclude that the QMJ factor, despite improving the return of some traditional factors, presents a statistically insignificant and negative risk premium. We posit that the quality measures used and / or restrictions in the Brazilian market may have affected these results, leaving room for improvement for future studies on the subject.

## 5. Conclusion

A quality company is one that is safe, profitable, and growing, according to Asness et al. (2019). Consistent with the international literature, we find

evidence that this characteristic requires higher adjusted prices for Brazilian investors. In line with others' results, we show that the companies of higher quality retain this characteristic in later periods, suggesting that quality is a persistent characteristic.

We find evidence that, in Brazil, quality companies have a risk-adjusted return higher than lower-quality companies. However, they continue to present significant alpha, different from zero, when controlled by the quality minus junk (QMJ) factor, in addition to the traditional factors. The QMJ factor, which is long on highest-quality companies and short on lowest-quality companies, has a positive risk-adjusted return (Sharpe Index), below 1. This presupposes that, in practice, we would possibly have a lower result due to cost transaction details.

We show that this format, of including several indicators as a unique characteristic, presents a robust result. When we control the main factors in the literature by the quality factor, we see an improvement in their intercepts. In addition, including the QMJ factor as a possible risk factor in traditional models improves the explanation, although part of the portfolios' returns remains unexplained by the model.

Finally, this study contributes to the literature, following the suggestion of [Chagas et al. \(2020\)](#). While those authors point out that a strategy based on investing in well-founded companies reduces the likelihood of bad returns; our study suggests that quality companies have higher risk-adjusted returns than others, and that this characteristic persists over time. We believe that this study helps to explain and analyze the performance of equity funds with active management, especially in management focused on buying quality companies.

## References

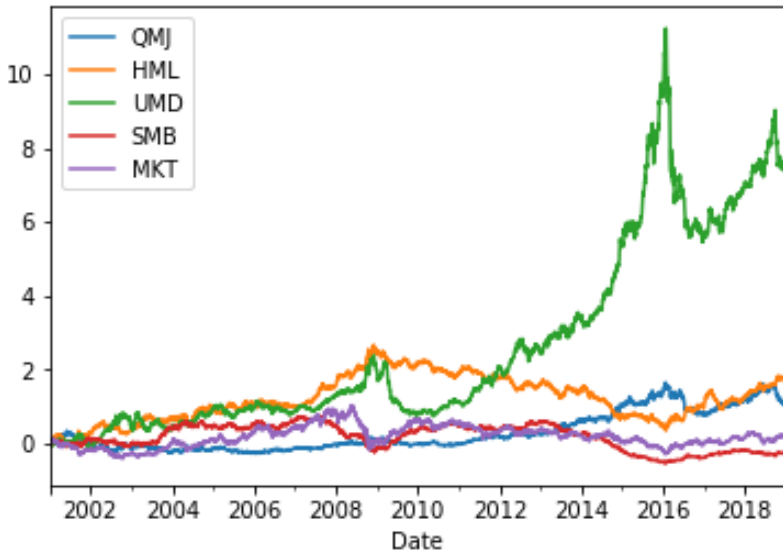
- Asness, C. S., Frazzini, A. and Pedersen, L. H. (2019). Quality minus junk, *Review of Accounting Studies* **24**(1): 34-112.
- Asness, C. S., Frazzini, A., Israel, R., Moskowitz, T. J. and Pedersen, L. H. (2018). Size matters, if you control your junk, *Journal of Financial Economics* **129**(3): 479-509.
- Basu, S. (1975). The information content of price-earnings ratios, *Financial Management* **4**(2): 53-64.
- Basu, S. (1983). The relationship between earnings' yield, market value and

- return for NYSE common stocks: Further evidence, *Journal of Financial Economics* **12**(1), 129-156.
- Banz, R. W. (1981). The relationship between return and market value of common stocks, *Journal of Financial Economics* **9**(1): 3-18.
- Carhart, M. M. (1997). On persistence in mutual fund performance, *Journal of Finance* **52**(1): 57-82.
- Chagas, L., Leal, R. and Roquete, R. (2020). Do good fundamentals generate alpha?, *Revista de Educação e Pesquisa em Contabilidade* **14**(2): 221-236.
- Fama, E. F. and French K. R. (1993). Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* **33**(1): 3-56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model, *Journal of Financial Economics* **116**(1): 1-22.
- Fama, E. F. and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests, *Journal of Political Economy*, **81**(3): 607-636.
- Frazzini, A., Kabiller, D. and Pedersen, L. H. (2018) Buffett's alpha, *Financial Analysts Journal* **74**(4):35-55.
- Frazzini, A. and Pedersen, L. H. (2014). Betting against beta, *Journal of Financial Economics* **111**(1): 1-25.
- George, T. J. and Hwang, C. A. (2010). Resolution of the distress risk and leverage puzzles in the cross section of stock returns, *Journal of Financial Economics* **96**(1): 56-79.
- Gordon, M. J. (1962). *The Investment, Financing, and Valuation of the Corporation*. Homewood, IL: RD Irwin.
- Graham, B. and Dodd, D. L. (1934). *Security Analysis*. New York: McGraw-Hill.
- Graham, B. (1973) *The Intelligent Investor*. Harpers & Row, New York. 4th rev. ed.
- Greenblatt, J. (2010). *The Little Book that Still Beats the Market* (Vol. 29). John Wiley & Sons.
- Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency, *Journal of Finance* **48**(1): 65-91.

- Jensen, M. C. (1968). The performance of mutual funds in the period 1945-1964, *Journal of Finance* **23**: 389–416.
- Markowitz, H. (1959). *Portfolio Selection: Efficient Diversification of Investments*. Yale University Press, New Haven.
- Martins, C. C. and Eid Jr, W. (2015). Pricing assets with Fama and French 5–Factor Model: A Brazilian market novelty, *Proceedings from the XV Encontro Brasileiro de Finanças*: 23-25.
- Minardi, A. M. A. F. and Sanvicente, A. Z. (2006). Análise da série histórica de prêmios pelo risco de mercado estimados pelo modelo de dividendos descontados, *Proceedings from the Encontro Brasileiro de Finanças*.
- Mohanram, P. S. (2005). Separating winners from losers among low book-to-market stocks using financial statement analysis, *Review of Accounting Studies* **10**: 133–170.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica* **55**(3): 703–708.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium, *Journal of Financial Economics* **108**(1): 1-28.
- Penman, S. H., Richardson, S. A. and Tuna, I. (2007). the book-to-price effect in stock returns: Accounting for leverage, *Journal of Accounting Research* **45**(2): 427–467.
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk, *Journal of Finance* **19**(3): 425-442.
- Stattman, D. (1980). Book values and stock returns, *The Chicago MBA: A Journal of Selected Papers* **4**(1): 25-45.
- Tobin, J. (1958). Estimation of relationships for limited dependent variables, *Econometrica* **31**: 24–36.
- Varga, G. and Brito, R. D. (2016). The cross-section of expected stock returns in Brazil, *Brazilian Review of Finance* **14**(2): 151-187.

## A. Figures and tables

**Figure A1**  
Cumulative results of the QMJ portfolio and main risk factors



**Table A1**  
**Statistical summary of data used in the quality metric**

|               | indicator | # quarters | average | sd      | min.     | 25%    | 50%    | 75%    | max.     |
|---------------|-----------|------------|---------|---------|----------|--------|--------|--------|----------|
| Profitability | GPOA      | 43,83      | 0,06    | 0,02    | 0,01     | 0,04   | 0,05   | 0,07   | 0,13     |
|               | ROE       | 45,68      | 2,20    | 22,67   | -63,54   | -6,11  | 3,96   | 13,20  | 47,93    |
|               | ROA       | 47,36      | -291,61 | 1318,27 | -7772,47 | -6,31  | 0,69   | 5,45   | 28,42    |
|               | CFOA      | 39,28      | 0,02    | 0,13    | -0,22    | -0,02  | 0,00   | 0,02   | 0,35     |
|               | GMAR      | 42,41      | 11,22   | 153,69  | -1245,23 | 22,80  | 31,51  | 37,30  | 55,19    |
|               | ACC       | 43,08      | 0,84%   | 0,48%   | 0,24%    | 0,53%  | 0,65%  | 1,10%  | 2,21%    |
| Growth        | ΔGPOA     | 26,77      | -0,11%  | 0,37%   | -1,03%   | -0,25% | -0,06% | 0,06%  | 0,59%    |
|               | ΔROE      | 26,95      | -25,09  | 102,12  | -244,95  | -21,35 | 19,12  | 24,94  | 42,92    |
|               | ΔROA      | 28,11      | 508,98  | 980,99  | -274,95  | -7,56  | 17,54  | 758,97 | 3350,38  |
|               | ΔCFOA     | 20,29      | 0,00    | 0,05    | -0,15    | -0,01  | 0,00   | 0,01   | 0,09     |
|               | ΔGMAR     | 25,83      | -1,45   | 23,51   | -157,15  | -0,51  | -0,10  | 0,56   | 77,66    |
|               |           |            |         |         |          |        |        |        |          |
| Safety        | -Beta     | 59,57      | -0,32   | 1,15    | -2,24    | -0,65  | -0,45  | -0,23  | 4,49     |
|               | LEV       | 51,19      | -0,37   | 0,27    | -1,29    | -0,47  | -0,27  | -0,21  | -0,12    |
|               | O-Score   | 45,54      | -14,96  | 59,16   | -359,38  | -0,05  | 2,11   | 2,69   | 7,18     |
|               | Z-Altmann | 29,09      | 1028,92 | 2228,33 | 9,69     | 168,98 | 391,01 | 716,97 | 12029,71 |
|               | ROE Vol   | 42,33      | 94,96   | 98,71   | 10,20    | 15,73  | 49,70  | 165,90 | 301,65   |
| PB            | Log PB    | 44,99      | 0,40    | 0,64    | -1,11    | -0,02  | 0,38   | 0,83   | 1,98     |

Note: we present average values for each variable. # quarters represents the number of average quarters of data found in each indicator; average represents the average value for each indicator; sd represents the average standard deviation for each indicator; min. represents the average minimum value for each indicator; 25% represents the 1st average quartile for each indicator; 50%, represents the average median for each indicator; 75%, represents the 3rd average quartile for each indicator; max. represents the average maximum value for each indicator.

**Table A2**  
**Persistence of quality variables**

| Brazilian stocks<br>2000-2018 | P1<br>(high) | P2   | P3   | P4   | P5    | P6    | P7    | P8    | P9    | P10<br>(low) | H –L | H –L<br>p-value |
|-------------------------------|--------------|------|------|------|-------|-------|-------|-------|-------|--------------|------|-----------------|
| Quality <sub>t</sub>          | 1.54         | 1.19 | 0.85 | 0.51 | 0.17  | –0.18 | –0.51 | –0.85 | –1.20 | –1.55        | 3.09 | $p < 0.001$     |
| Quality <sub>t+3</sub>        | 1.42         | 1.03 | 0.72 | 0.43 | 0.13  | –0.16 | –0.46 | –0.74 | –1.00 | –1.38        | 2.79 | $p < 0.001$     |
| Quality <sub>t+6</sub>        | 1.35         | 0.96 | 0.70 | 0.40 | 0.10  | –0.17 | –0.44 | –0.70 | –0.97 | –1.28        | 2.63 | $p < 0.001$     |
| Quality <sub>t+12</sub>       | 1.26         | 0.84 | 0.62 | 0.34 | 0.06  | –0.15 | –0.42 | –0.63 | –0.88 | –1.09        | 2.35 | $p < 0.001$     |
| Quality <sub>t+24</sub>       | 1.02         | 0.68 | 0.52 | 0.28 | –0.04 | –0.08 | –0.32 | –0.51 | –0.71 | –0.84        | 1.85 | $p < 0.001$     |
| Profitability <sub>t</sub>    | 1.54         | 1.19 | 0.85 | 0.51 | 0.16  | –0.18 | –0.52 | –0.86 | –1.20 | –1.55        | 3.09 | $p < 0.001$     |
| Profitability <sub>t+3</sub>  | 1.42         | 1.06 | 0.74 | 0.43 | 0.08  | –0.21 | –0.36 | –0.70 | –1.04 | –1.40        | 2.83 | $p < 0.001$     |
| Profitability <sub>t+6</sub>  | 1.38         | 1.02 | 0.73 | 0.40 | 0.04  | –0.18 | –0.37 | –0.63 | –0.97 | –1.34        | 2.73 | $p < 0.001$     |
| Profitability <sub>t+12</sub> | 1.35         | 0.94 | 0.66 | 0.36 | 0.02  | –0.21 | –0.33 | –0.57 | –0.84 | –1.21        | 2.56 | $p < 0.001$     |
| Profitability <sub>t+24</sub> | 1.25         | 0.87 | 0.59 | 0.30 | 0.06  | –0.16 | –0.28 | –0.50 | –0.76 | –1.04        | 2.29 | $p < 0.001$     |
| Growth <sub>t</sub>           | 1.54         | 1.18 | 0.83 | 0.49 | 0.15  | –0.18 | –0.52 | –0.86 | –1.19 | –1.54        | 3.08 | $p < 0.001$     |
| Growth <sub>t+3</sub>         | 1.24         | 0.82 | 0.62 | 0.38 | 0.16  | –0.12 | –0.33 | –0.62 | –0.93 | –1.30        | 2.54 | $p < 0.001$     |
| Growth <sub>t+6</sub>         | 1.02         | 0.72 | 0.54 | 0.34 | 0.12  | –0.05 | –0.27 | –0.51 | –0.82 | –1.19        | 2.22 | $p < 0.001$     |
| Growth <sub>t+12</sub>        | 0.69         | 0.53 | 0.45 | 0.30 | 0.12  | –0.02 | –0.20 | –0.37 | –0.58 | –1.02        | 1.71 | $p < 0.001$     |
| Growth <sub>t+24</sub>        | 0.15         | 0.31 | 0.32 | 0.32 | 0.03  | 0.11  | –0.02 | –0.23 | –0.40 | –0.76        | 0.91 | $p < 0.001$     |
| Safety <sub>t</sub>           | 1.54         | 1.19 | 0.85 | 0.51 | 0.17  | –0.18 | –0.51 | –0.86 | –1.20 | –1.55        | 3.09 | $p < 0.001$     |
| Safety <sub>t+3</sub>         | 1.46         | 1.09 | 0.80 | 0.46 | 0.13  | –0.17 | –0.48 | –0.78 | –1.10 | –1.47        | 2.93 | $p < 0.001$     |
| Safety <sub>t+6</sub>         | 1.42         | 1.02 | 0.78 | 0.44 | 0.11  | –0.14 | –0.48 | –0.75 | –1.08 | –1.43        | 2.85 | $p < 0.001$     |
| Safety <sub>t+12</sub>        | 1.35         | 0.93 | 0.75 | 0.37 | 0.07  | –0.15 | –0.44 | –0.71 | –1.01 | –1.37        | 2.71 | $p < 0.001$     |
| Safety <sub>t+24</sub>        | 1.15         | 0.81 | 0.69 | 0.28 | 0.03  | –0.19 | –0.42 | –0.61 | –0.87 | –1.20        | 2.35 | $p < 0.001$     |

Note: This table shows the average quality factor z-score. For each month, we rank the shares based on their quality score. Then we rank them in one of the 10 portfolios in ascending order. The table shows the average quality score on the date the portfolio was formed (t) and in subsequent months (t + 3, t + 6, t + 12 and t + 24). Our sample runs from January 2000 to December 2018. We adjust standard errors for heteroskedasticity and autocorrelation with a 6-month lag (Newey and West, 1987). Variables with a 5% statistical significance are in bold.

**Table A3**  
**Price regression against quality score for each equation**

| Brazilian stocks<br>2000-2018   | (3)                      | (4)                      | (5)                     | (6)                       | (7)                       |
|---------------------------------|--------------------------|--------------------------|-------------------------|---------------------------|---------------------------|
| Quality                         | <b>0.1741</b><br>(18.55) |                          |                         |                           |                           |
| Profitability                   |                          | <b>0.3906</b><br>(55.22) |                         |                           | <b>0.4208</b><br>(37.02)  |
| Growth                          |                          |                          | <b>0.0665</b><br>(5.73) |                           | <b>-0.0466</b><br>(-4.09) |
| Safety                          |                          |                          |                         | <b>-0.0309</b><br>(-2.31) | <b>-0.1168</b><br>(-8.68) |
| avg. of adjusted R <sup>2</sup> | 2.70%                    | 9.10%                    | -0.20%                  | -0.50%                    | 8.95%                     |
| avg. obs                        | 187                      | 185.26                   | 133.26                  | 187                       | 133.26                    |

Note: This table shows the [Fama and MacBeth \(1973\)](#) regression coefficient, that is, the average betas of each company. The dependent variable is the log of the equity value price of the shares. The columns represent each of the equations presented previously. The independent variable is the quality z-score, and its individual components. T-statistics are in parentheses, and variables with a 5% statistical significance are in bold.

**Table A4**  
**Quality Minus Junk: Correlation between returns of the main factors**

| Corr.         | QMJ   | Profitability | Growth | Safety | UMD   | HML   | MKT   | SMB   |
|---------------|-------|---------------|--------|--------|-------|-------|-------|-------|
| QMJ           | 1.00  | 0.75          | 0.18   | 0.68   | 0.24  | -0.27 | -0.28 | -0.09 |
| Profitability | 0.75  | 1.00          | 0.05   | 0.53   | 0.22  | -0.33 | -0.20 | -0.10 |
| Growth        | 0.18  | 0.05          | 1.00   | -0.05  | 0.05  | 0.01  | -0.11 | 0.00  |
| Safety        | 0.68  | 0.53          | -0.05  | 1.00   | 0.15  | -0.22 | -0.29 | -0.06 |
| UMD           | 0.24  | 0.22          | 0.05   | 0.15   | 1.00  | -0.20 | -0.04 | -0.33 |
| HML           | -0.27 | -0.33         | 0.01   | -0.22  | -0.20 | 1.00  | 0.22  | 0.16  |
| MKT           | -0.28 | -0.20         | -0.11  | -0.29  | -0.04 | 0.22  | 1.00  | -0.33 |
| SMB           | -0.09 | -0.10         | 0.00   | -0.06  | -0.33 | 0.16  | -0.33 | 1.00  |

Note: This table reports the correlation of the returns of the main factors. We create the quality minus junk (QMJ) factor with the purchase and sale of four portfolios, formed according to their size and quality. We separate companies into small and large, based on their market value, and for quality, based on the score created previously. The weights of each asset within the portfolios are based on market size. The factors UMD, HML, MKT, and SMB were collected from NEFIN-USP.

**Table A5**  
**Average annualized return of the regression of the percentile return with risk factors**

|                      | P1 (low)       | P2             | P3                    | P4                    | P5                    | P6             | P7             | P8                    | P9                    | P10 (high)            |
|----------------------|----------------|----------------|-----------------------|-----------------------|-----------------------|----------------|----------------|-----------------------|-----------------------|-----------------------|
| Excess Return        | 0.09<br>(1.07) | 0.09<br>(1.14) | 0.16<br>(1.97)        | 0.14<br>(1.96)        | <b>0.14</b><br>(2.47) | 0.10<br>(1.50) | 0.08<br>(1.06) | 0.11<br>(1.92)        | <b>0.17</b><br>(2.98) | <b>0.19</b><br>(3.33) |
| Market Factor Alpha  | 0.04<br>(0.87) | 0.04<br>(0.97) | 0.11<br>(2.06)        | <b>0.10</b><br>(2.05) | <b>0.11</b><br>(3.00) | 0.06<br>(1.49) | 0.04<br>(0.75) | <b>0.08</b><br>(2.08) | <b>0.14</b><br>(3.69) | <b>0.15</b><br>(4.33) |
| 3-Factor Alpha       | 0.02<br>(0.51) | 0.03<br>(0.66) | 0.11<br>(1.94)        | 0.10<br>(1.97)        | <b>0.11</b><br>(3.11) | 0.06<br>(1.43) | 0.05<br>(0.86) | <b>0.08</b><br>(2.18) | <b>0.14</b><br>(3.90) | <b>0.16</b><br>(4.53) |
| 4 Factor Alpha       | 0.05<br>(1.06) | 0.04<br>(0.95) | <b>0.11</b><br>(2.07) | 0.09<br>(1.81)        | <b>0.11</b><br>(2.93) | 0.05<br>(1.24) | 0.03<br>(0.60) | 0.08<br>(1.98)        | 0.14<br>(3.70)        | <b>0.14</b><br>(4.10) |
| 4 Factor + Qmj Alpha | 0.06<br>(1.36) | 0.05<br>(1.20) | <b>0.12</b><br>(2.33) | <b>0.11</b><br>(2.15) | <b>0.11</b><br>(3.23) | 0.05<br>(1.27) | 0.03<br>(0.54) | 0.07<br>(1.80)        | <b>0.13</b><br>(3.55) | <b>0.13</b><br>(3.92) |
| Beta                 | 1.10           | 1.06           | 0.97                  | 0.94                  | 0.79                  | 0.88           | 0.88           | 0.76                  | 0.82                  | 0.77                  |

Note: For each month, we rank companies based on the quality z-score and place them in one of the 10 portfolios. The weights of each asset are based on market size. Alpha is the time series regression intercept. Alphas are annualized, and t-statistics are shown below the coefficients. Coefficients that are statistically significant at 5% are in bold.

**Table A6**  
**Regression of the return of QMJ factor with risk factors**

|                         | dependent variable       |                          |                        |                         |                          |                          |
|-------------------------|--------------------------|--------------------------|------------------------|-------------------------|--------------------------|--------------------------|
|                         | SMB                      | SMB                      | HML                    | HML                     | UMD                      | UMD                      |
| Excess Return           | <b>-1.5%</b><br>(-6.78)  | <b>-1.5%</b><br>(-6.78)  | <b>5.7%</b><br>(27.68) | <b>5.7%</b><br>(27.68)  | <b>12.2%</b><br>(50.5)   | <b>12.2%</b><br>(50.5)   |
| Alpha                   | 0.02<br>(0.73)           | 0.03<br>(0.84)           | 0.06<br>(1.86)         | <b>0.07</b><br>(2.0)    | <b>0.14</b><br>(3.45)    | <b>0.13</b><br>(3.24)    |
| MKT                     | <b>-0.18</b><br>(-10.89) | <b>-0.19</b><br>(-11.07) | <b>0.14</b><br>(7.99)  | <b>0.13</b><br>(7.45)   | <b>-0.12</b><br>(-5.81)  | <b>-0.11</b><br>(-5.1)   |
| SMB                     |                          |                          | <b>0.25</b><br>(8.69)  | <b>0.24</b><br>(8.33)   | <b>-0.38</b><br>(-12.37) | <b>-0.37</b><br>(-12.07) |
| HML                     | <b>0.25</b><br>(8.93)    | <b>0.24</b><br>(8.4)     |                        |                         | <b>-0.03</b><br>(-0.84)  | -0.01<br>(-0.34)         |
| UMD                     | <b>-0.28</b><br>(-10.55) | <b>-0.27</b><br>(-10.55) | -0.02<br>(-0.83)       | -0.01<br>(-0.34)        |                          | -0.06<br>(-1.95)         |
| QMJ                     |                          | <b>-0.07</b><br>(-2.79)  |                        | <b>-0.10</b><br>(-5.26) |                          | <b>0.12</b><br>(4.76)    |
| information ratio       | 0.18                     | 0.20                     | 0.49                   | 0.53                    | 0.93                     | 0.88                     |
| adjusted R <sup>2</sup> | 21.18%                   | 21.60%                   | 10.76%                 | 11.89%                  | 12.88%                   | 14.13%                   |

Note: Alpha is the time series regression intercept. The explanatory variables are market return (MKT), size (SMB), value (HML), and momentum (UMD). Both returns and alphas are annualized, and t-statistics are shown below the coefficients. Coefficients that are statistically significant at 5% are in bold. The information ratio is the result of the alpha of the four-factor risk model, divided by the standard deviation of the regression residuals.

**Table A7**  
**Fama and MacBeth (1973) regression of the main asset pricing models**

|             | QMJ Factor         | Carhart 4 Factor   | Fama French 3 Factor | CAPM             |
|-------------|--------------------|--------------------|----------------------|------------------|
| MKT         | 0.0006<br>(1.55)   | 0.0006<br>(1.84)   | 0.0006<br>(1.71)     | 0.0004<br>(0.33) |
| SMB         | -0.0002<br>(-0.17) | -0.0001<br>(-0.14) | 0.0000<br>(-0.04)    |                  |
| HML         | 0.0019<br>(1.42)   | 0.0020<br>(1.43)   | 0.0026<br>(1.95)     |                  |
| UMD         | -0.0018<br>(-0.65) | -0.0018<br>(-0.65) |                      |                  |
| QMJ         | -0.0009<br>(-0.55) |                    |                      |                  |
| Adjusted R2 | 0.81%              | 0.81%              | 0.79%                | 0.67%            |

Note: Both returns and alphas are annualized, and t-statistics are below the coefficients. Coefficients that are statistically significant at 5% are in bold.